

MINOR PROJECT-5 REPORT

CNN Image Classification using MNIST and Streamlit Web Application

Submitted By

Name: Keshav V Marathe

Course: Machine Learning

Batch: June-2026

Date of Submission: 26 July 2026

Dataset Used:

MNIST Handwritten Digit Dataset

Streamlit Application Topic:

Handwritten Digit Recognition using CNN

1. Objective

The main objective of this project is to understand the implementation of Convolutional Neural Networks (CNN) for image classification and to develop a simple Streamlit web application for predicting handwritten digits.

In Part A, the MNIST handwritten digit dataset is used to train a CNN model. The images are preprocessed, the model is trained, and its performance is evaluated using the test dataset. The prediction results are analyzed by comparing the actual and predicted digit labels.

In Part B, the trained CNN model is integrated with a Streamlit application. The application allows users to upload an image of a handwritten digit and predicts the corresponding digit along with the confidence score. This provides a simple and interactive interface for testing the trained model.

2. CNN Workflow

2.1 Dataset Description

For this project, the **MNIST (Modified National Institute of Standards and Technology)** dataset was selected. It is a standard image classification dataset containing handwritten digits from **0 to 9**. The dataset consists of grayscale images of size **28 × 28 pixels**.

The dataset contains:

- Training Images: 60,000
- Training Labels: 60,000
- Testing Images: 10,000
- Testing Labels: 10,000

The ten output classes are:

0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

2.2 Data Preprocessing

Before training the CNN model, the dataset was preprocessed to improve model performance.

The preprocessing steps performed were:

- Loaded the MNIST dataset from TensorFlow.
- Normalized pixel values from **0–255** to **0–1** by dividing each pixel by 255.
- Reshaped the images into **28 × 28 × 1** format so that they could be used as input for the CNN model.

2.3 CNN Model Architecture

A Convolutional Neural Network (CNN) was created using TensorFlow Keras.

The model consists of:

- Input Layer ($28 \times 28 \times 1$)
- Convolution Layer (32 filters, 3×3)
- Max Pooling Layer
- Convolution Layer (64 filters, 3×3)
- Max Pooling Layer
- Flatten Layer
- Dense Layer (64 neurons)
- Output Layer (10 neurons with Softmax activation)

The model was compiled using:

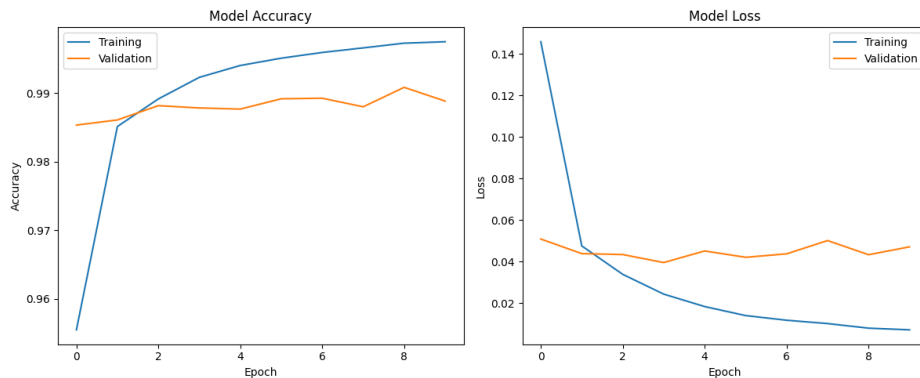
- Optimizer: Adam
- Loss Function: Sparse Categorical Crossentropy
- Evaluation Metric: Accuracy

2.4 Model Training

The CNN model was trained using the training dataset with the following parameters:

- Epochs: 10
- Batch Size: 32
- Validation Split: 20%

During training, both training accuracy and validation accuracy improved gradually, while the loss decreased with each epoch.



2.5 Model Evaluation

After training, the model was evaluated using the test dataset.

The trained model achieved a test accuracy of approximately **98.85%**, which indicates that it can classify handwritten digits with high accuracy.

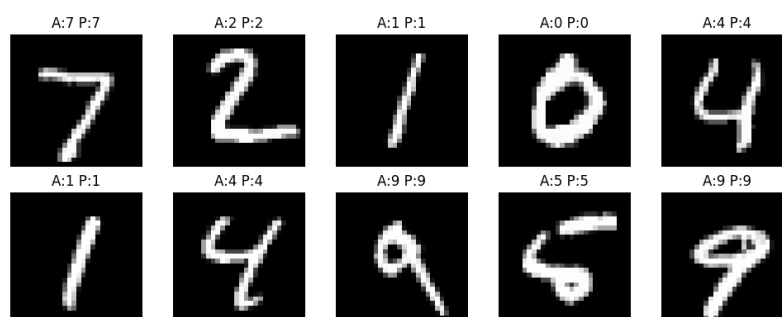
The trained model was saved as:

mnist_cnn_model.keras

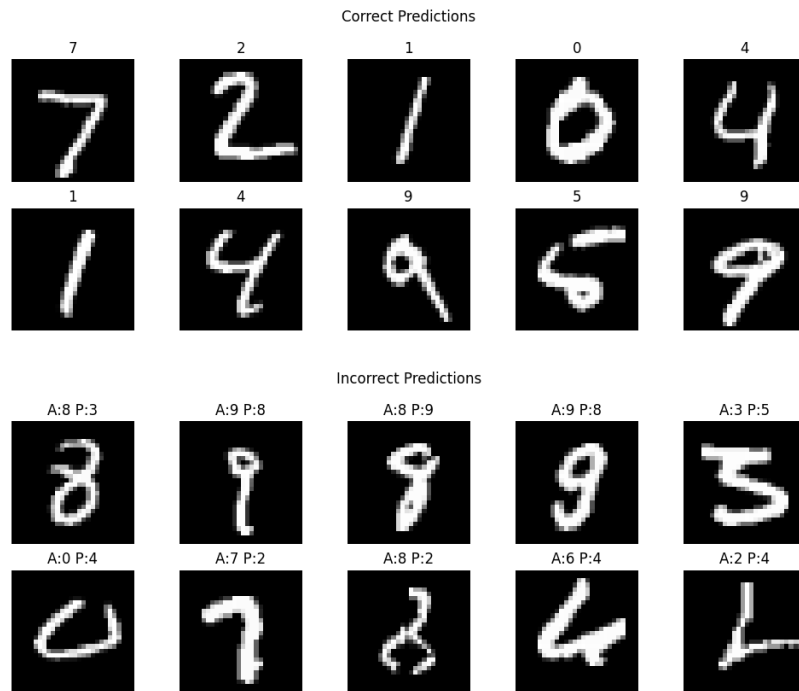
2.6 Prediction Results

The trained CNN model was tested using unseen images from the test dataset.

The project displays:



Actual vs Predicted Digits



These prediction results help in understanding the performance of the CNN model and identifying cases where the model makes incorrect predictions.

3. Streamlit Application

3.1 Application Overview

A Streamlit web application was developed to provide a simple interface for testing the trained CNN model. The application allows users to upload an image of a handwritten digit and predicts the digit using the saved CNN model.

The trained model (`mnist_cnn_model.keras`) is loaded into the application, enabling users to test the model without retraining it.



Streamlit Home Page

3.2 User Input

The application accepts an image file from the user through a file uploader.

Supported image formats are:

- PNG (.png)
- JPG (.jpg)
- JPEG (.jpeg)

The uploaded image is displayed on the screen before prediction.

3.3 Image Preprocessing

Before prediction, the uploaded image undergoes the following preprocessing steps:

- Converted to grayscale.
- Image colors are inverted to match the MNIST dataset format.
- Resized to **28 × 28 pixels**.
- Pixel values are normalized between **0 and 1**.
- Reshaped into **28 × 28 × 1** format for CNN input.

These preprocessing steps ensure that the uploaded image matches the format used during model training.

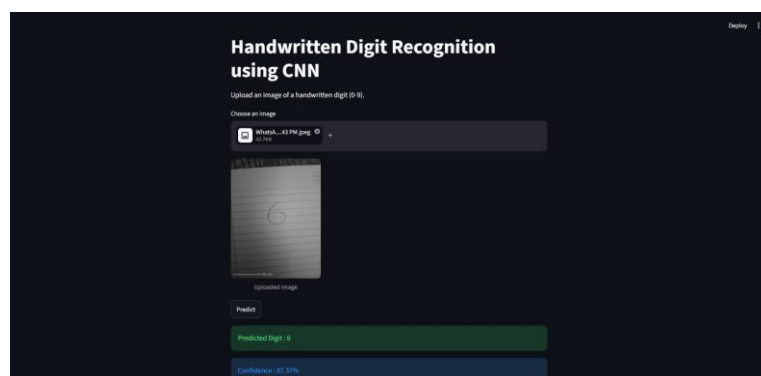
3.4 Prediction

When the **Predict** button is clicked, the application sends the processed image to the trained CNN model.

The application displays:

- Predicted Digit
- Prediction Confidence (%)

This provides users with both the classification result and the confidence level of the prediction.



Prediction Result

3.5 Features of the Application

The Streamlit application provides the following features:

- Simple and user-friendly interface.
- Image upload support.
- Automatic image preprocessing.
- Real-time digit prediction.
- Displays prediction confidence.
- Uses the previously trained CNN model without retraining.

The application runs successfully in a web browser and provides a quick and interactive way to test handwritten digit recognition.

4. Results

The CNN model was successfully trained using the MNIST handwritten digit dataset. During training, both the training accuracy and validation accuracy improved steadily, while the training and validation loss decreased over each epoch. This indicates that the model learned the image features effectively.

After evaluating the model on the test dataset, it achieved a **test accuracy of 98.85%**, demonstrating excellent performance in classifying handwritten digits.

The Streamlit application was also developed successfully. It correctly loads the saved CNN model and allows users to upload handwritten digit images for prediction. After preprocessing the uploaded image, the application predicts the digit and displays the confidence score, making it easy for users to interact with the trained model.

Observations

- The CNN model achieved high classification accuracy on the MNIST dataset.
- Most handwritten digits were predicted correctly.
- The Streamlit application provided fast and accurate predictions.
- The integration of the trained CNN model with the Streamlit interface was successful.
- The project demonstrated the complete workflow from model training to real-time prediction using a web application.